

HUY NGUYEN QUOC GIA

Full Stack Software Engineer (AI Focus)

0975440503 | nqghuy1202@gmail.com | Ho Chi Minh City | [Linkedin](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Full Stack Software Engineer (AI Focus) with 2 years of production ERP experience and a self-built, production-deployed RAG assistant. Designed an on-premise agentic LLM pipeline that lets external ERP customers query and modify data in natural language, with confirmation and audit logging on every write or delete. Now used by 20+ customers daily, cutting support questions by 30%. Proficient in Python, TypeScript, React and REST APIs through personal projects, with Git and Linux from daily production work.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, PL/SQL

AI/LLM: RAG pipeline design, Ollama, Oracle AI Database 26ai, bge-m3 embeddings, MCP servers, BMAD-METHOD, Claude Code

Frontend: React, TypeScript, Vite, Tailwind CSS

Backend & APIs: Django, Node.js, Oracle APEX/PL/SQL, REST API design

Infrastructure & Tools: Git, Linux (server administration), nginx, MySQL, Oracle Database

PROFESSIONAL EXPERIENCE

Gia Cat Technology Solutions Co., Ltd – Full Stack Software Engineer

07/2024 – 08/2026

ERP consulting and implementation for construction, manufacturing and other industries

On-Premise AI Assistant for ERP

- **Technologies:** Ollama (qwen2.5:3b-instruct), bge-m3 embeddings, Oracle AI Database 26ai, HNSW vector indexing, PL/SQL, JavaScript, TypeScript, HTML, CSS
- **Description:** Self-initiated, on-premise natural-language assistant embedded in the company's Oracle APEX ERP, letting external ERP customers query and modify data in plain language instead of contacting support.
- Designed an agentic dual-pipeline architecture where the LLM classifies user intent and routes to either a RAG pipeline (bge-m3 embeddings, HNSW vector indexing) for lookups, or a guarded Natural Language to SQL flow requiring confirmation, audit logging, and rollback for modifications.
- Traced a severe latency issue on CPU-only infrastructure to a security token breaking Ollama's prefix cache, using packet capture. Redesigned the flow into a single-call architecture to eliminate it.
- Evaluated the Natural Language to SQL flow against 12 Vietnamese queries (100% correct) and 4 SQL-injection attempts.
- Used Claude Code with the BMAD-METHOD multi-agent workflow to take the project from concept to working MVP.
- **Achievements:** Cut average response time from 153 seconds to about 40 seconds. Adopted by 20+ external ERP customers for daily use. Cut data lookup time by 20% and reduced inbound support-question volume by 30%.

Real-time Chat System

- **Technologies:** Node.js, Server-Sent Events (SSE), HMAC-SHA256 token authentication, vanilla JavaScript, HTML, CSS
- **Description:** A Node.js middleware layer adding real-time push messaging and notifications to an Oracle APEX ERP with no native server-push capability.
- Diagnosed and fixed two production bugs: a cross-schema message-delivery collision and a notification-badge desync. Flagged an unauthenticated broadcast endpoint as a security risk.
- Built both the backend push service and the front-end chat interface, with no separate frontend team involved.
- **Achievements:** Deployed to production, used by 30+ external ERP customers daily. Designed a SharedWorker architecture that cut concurrent SSE connections by about 90%, eliminating a prior connection-limit issue.

PROJECTS

Heart Risk Estimator

GitHub: github.com/nqghuy1202/heart_risk_estimator

- **Technologies:** Python, Django, scikit-learn, pandas, joblib, React, TypeScript, Vite
- **Description:** A public web app that predicts heart disease risk from 13 clinical measurements, pairing a Python ML backend with a schema-driven React frontend. Personal project, open-source on GitHub.
- Built the Django backend as a JSON REST API, with a single schema endpoint that drives how the React frontend renders and validates its form – nothing hardcoded on the client.
- Trained and tuned an AdaBoost classifier in scikit-learn, manually sweeping the ensemble size to find the smallest model that held test accuracy. Covered the frontend with 15 automated integration tests (Vitest, React Testing Library).
- **Achievements:** Reached 87.8% accuracy and 92.9% recall on a 205-record held-out test set, against a 74.6% baseline.

EDUCATION

Bachelor of Information Technology – Data Science

Graduated 2025

Ho Chi Minh City University of Industry and Trade (HUIT) | GPA: 3.36/4.00